

■ AI SUMMIT

Minutes Matter

Fielding edge AI for personnel recovery

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ELAPSED

00:00

A pilot is down.

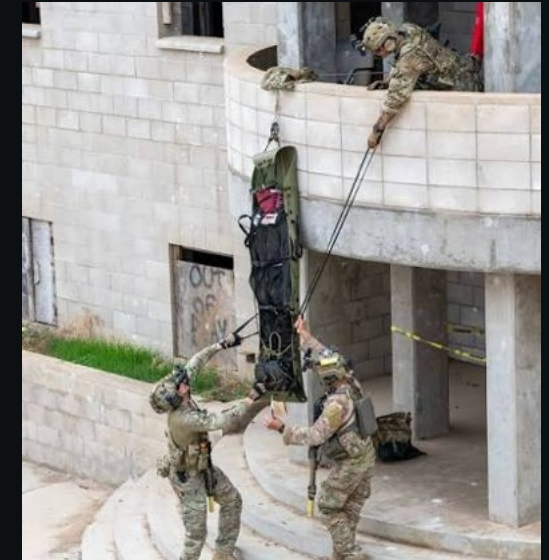
■ TWO VIEWPOINTS. ONE REFERENCE



FROM THE AIR: ROOFTOPS



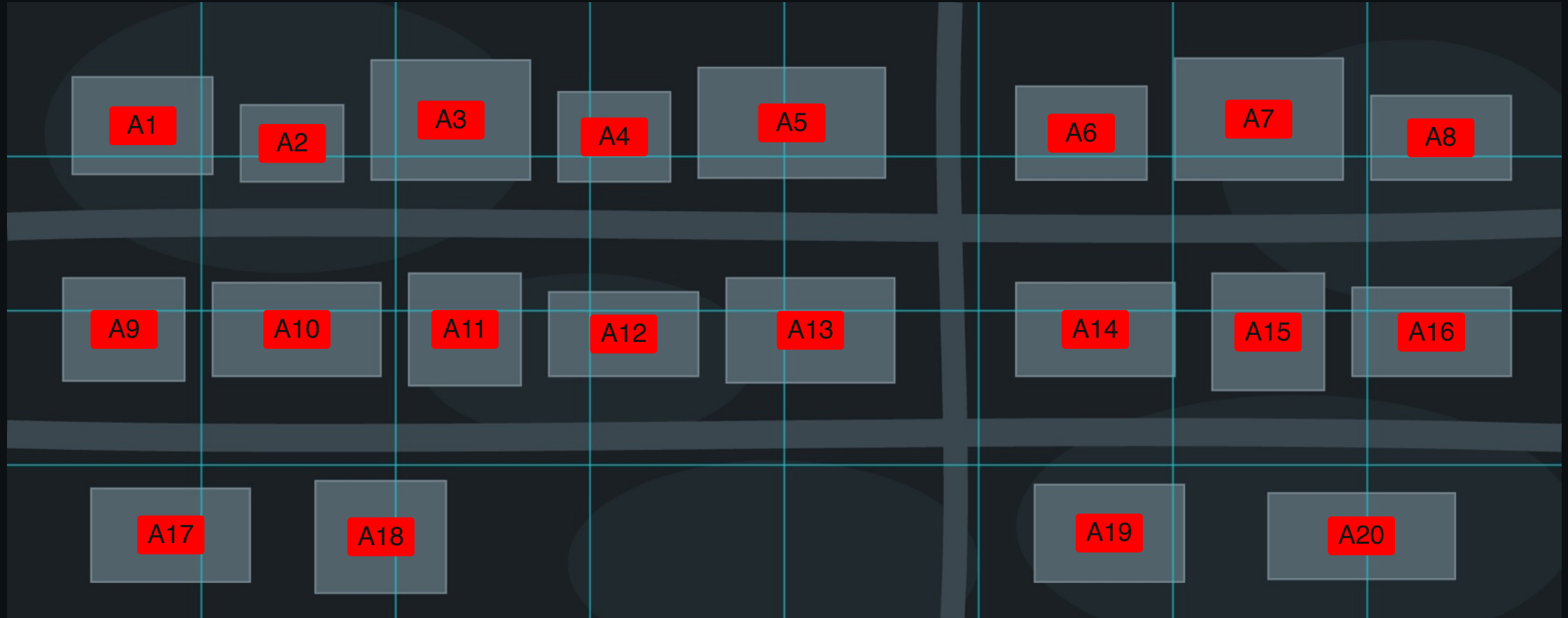
Our guy is in 205.



ON THE GROUND:
DOORS AND WALLS

🔍 FIND 🔒 SECURE 📋 TREAT 🏠 RECOVER

■ BUILT BY HAND. UNDER THE CLOCK



Now do that 100 more times. *By hand. On a phone.*

■ BUILT IN TAK. OUT ON THE EDGE



SMALL SCREENS

OLDER HARDWARE

NO CONNECTIVITY

LEGACY SOFTWARE

■ HEY, CAN YOU DO THIS?

Build a TAK plugin that makes GRGs faster

...with AI that detects and labels buildings

...running fully offline

...on older Android phones and low-spec Windows laptops

...with full interoperability to TAK, OGC, and other protocol standards

...meeting cyber compliance

...in ~~8~~ 7 6 weeks, with 6 people

The real requirement: give operators back the time they need to prepare for the rescue.

■ DAY ONE

- ☒ Assemble a team of switched-on people
- ☒ Buy S20 and S23 phones
- ☒ Buy Windows laptops
- ☒ Open Claude

■ MECHANICAL GAP

THE PROBLEM NEEDED

Native Android (Kotlin)

WPF on .NET 4.x

TAK plugin internals

A CV training pipeline

YOLO, YOLT, Mask R-CNN, U-Net

Edge runtimes (PyTorch, ONNX)

A custom .NET-to-Java JNI bridge

Complex path traversal math

To be solved

WHAT WE HAD

Hadn't touched it in forever

Never done it (what, no ARM support?)

Learning as we went

Never set one up

"Carpe diem?"

Never had to care before

Didn't know how to write one

Not our day job

Six people willing to figure it out

■ SO...IS IT A BUILDING



GARAGE



GAZEBO



TENT



■ ELUSIVE BUILDING MIMICS



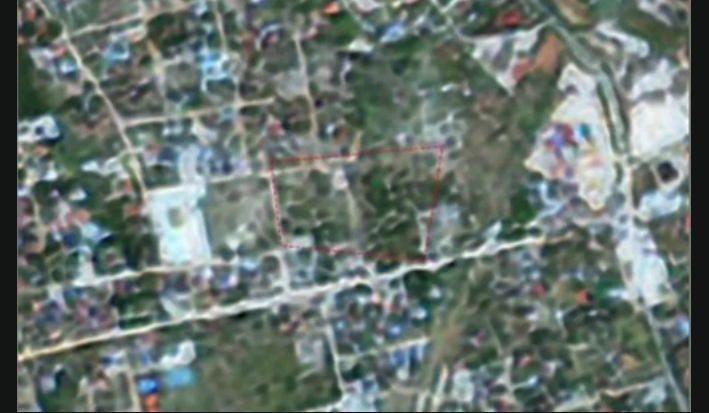
■ ANY IMAGERY. ON A PHONE. OFFLINE

AI BAKE-OFF

YOLO · YOLT · U-Net · Mask R-CNN,
plus various methods of preprocessing



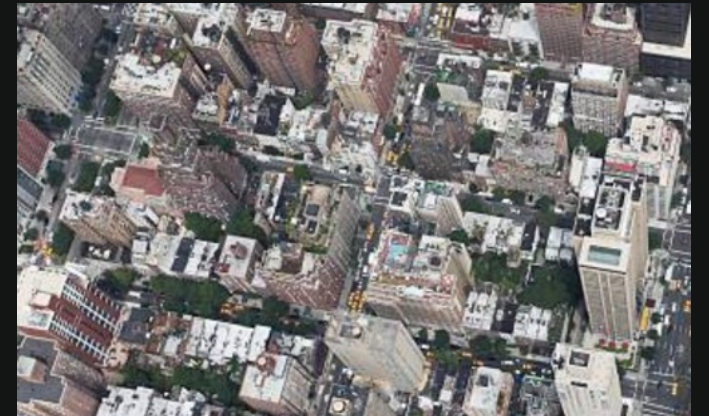
LOCATION



RESOLUTION



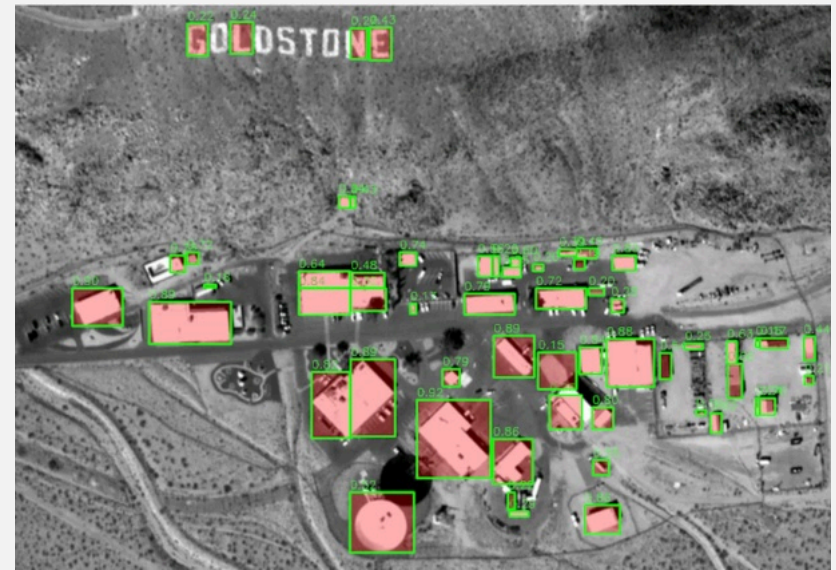
COLOR



ANGLE



YOLO



YOLT



MASK R-CNN



U-NET

■ HOW SHOULD LABELS READ?



■ HOW FAST CAN AN OPERATOR GET A GRG?

01



Select an area

Pick the area of interest on the map.

02



Auto-create the GRG

Detected, gridded, and labeled on the device in a couple of minutes, fully offline.

03



Refine it

- Recolor
- Swipe to relabel in order
- Edit, move, or delete labels
- Automatic relabeling

■ LESS LEGWORK. MORE MISSION

■ AI HANDLED

- TAK interoperability
- Geo math
- Map tiles to KMZ
- Imagery preprocessing
- Sliding window YOLT implementation
- Cross-section deduplication
- .NET-to-Java JNI bridge
- Training pipeline
- Lots of path traversal math
- Android, Kotlin, and WPF after years away

■ WE ENGINEERED

- What is a building, anyway?
- Any imagery, on the phone in their hand
- How labels should read
- A GRG in a couple of minutes
- Fixing it fast when the AI gets it wrong
- 100% buildings labeled all the time

■ MISSION-CRITICAL SOFTWARE HAS TO WORK

When a GRG is wrong, a team ends up at the wrong door.

It has to work.

It has to work well.

In their hands now. Not after another test cycle.

AI doesn't ship spaghetti. **Developers do.**

■ WHAT'S IN THEIR HANDS

- ✓ Buildings detected automatically, on any imagery
- ✓ Labels operators can actually follow
- ✓ End-to-end GRG workflows on Android and Windows
- ✓ Fully offline, on the devices they already carry
- ✓ Security standards and operational acceptance testing met
- ✓ ...all in 6 weeks

and it's in use

■ WHAT IT MEANS FOR THE MISSION

8+ hrs

by hand for a large urban GRG

➔ **a couple of minutes**

~30 min

back on the average smaller
GRG

Complete

not cut short: no more
simplified GRGs when time
runs out

More time

goes back to preparing for the
rescue

ELAPSED

00:00

Time is the mission.

Speed to field is a capability, not a nice-to-have.

■ ASK OF THIS ROOM

01 Start from the mission, not the tech.

02 Hand AI the mechanics. Focus people on the mission.

03 Own the quality bar. Then move fast.

04 Do more with less.

05 Don't be limited. Get creative with AI.

■ ASK FROM THE ROOM

01 Work-life balance when human time is a premium.

02 More, better, faster models on government systems.

■ MAXIMIZE THE MOMENT

Thank you

